

Simplifying Scalable Subspace Clustering and Its Multi-View Extension by Anchor-to-Sample Kernel

Zhoumin Lu¹, Feiping Nie¹, Senior Member, IEEE, Linru Ma¹, Rong Wang¹, and Xuelong Li², Fellow, IEEE

Abstract—As we all known, sparse subspace learning can provide good input for spectral clustering, thereby producing high-quality cluster partitioning. However, it employs complete samples as the dictionary for representation learning, resulting in non-negligible computational costs. Therefore, replacing the complete samples with representative ones (anchors) as the dictionary has become a more popular choice, giving rise to a series of related works. Unfortunately, although these works are linear with respect to the number of samples, they are often quadratic or even cubic with respect to the number of anchors. In this paper, we derive a simpler problem to replace the original scalable subspace clustering, whose properties are utilized. This new problem is linear with respect to both the number of samples and anchors, further enhancing scalability and providing more efficient operations. Furthermore, thanks to the new problem formulation, we can adopt a separate fusion strategy for multi-view extensions. This strategy can better measure the inter-view difference and avoid alternate optimization, so as to achieve more robust and efficient multi-view clustering. Finally, comprehensive experiments demonstrate that our methods not only significantly reduce time overhead but also exhibit superior performance.

Index Terms—Subspace clustering, scalable clustering, self-expression learning, graph learning, multi-view extension.

I. INTRODUCTION

AS a cornerstone of machine learning, clustering has been greatly developed, which groups data points with unknown labels. Among many classic algorithms, spectral clustering is impressive due to its excellent performance. It takes a graph as input and then regenerates feature representations [1] or directly performs subgraph segmentation [2].

Received 30 September 2024; revised 27 May 2025; accepted 16 July 2025. Date of publication 1 August 2025; date of current version 11 August 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62236001, Grant 62276212, and Grant 62176212. The associate editor coordinating the review of this article and approving it for publication was Dr. Zhengming Ding. (Corresponding authors: Feiping Nie; Linru Ma.)

Zhoumin Lu and Feiping Nie are with the School of Computer Science, School of Artificial Intelligence, Optics and ElectroNics (iOPEN), and the Key Laboratory of Intelligent Interaction and Applications (Ministry of Industry and Information Technology), Northwestern Polytechnical University, Xi'an 710072, China (e-mail: walker.zhoumin.lu@gmail.com; feipingnie@gmail.com).

Linru Ma is with the Institute of Data and Intelligence, Beijing 100071, China (e-mail: malinru@163.com).

Rong Wang is with the School of Artificial Intelligence, Optics and ElectroNics (iOPEN), Northwestern Polytechnical University, Xi'an 710072, China (e-mail: wangrong07@tsinghua.org.cn).

Xuelong Li is with the Institute of Artificial Intelligence (TeleAI) of China Telecom, Shanghai 200232, China (e-mail: xuelong_li@ieee.org).

Digital Object Identifier 10.1109/TIP.2025.3593057

If a graph cannot well describe the correlation between samples, spectral clustering will naturally not produce the expected results. Undoubtedly, graph learning plays a crucial role in spectral clustering, which has led to many algorithms being proposed. Given sample features, USPEC [3] employs a hybrid representative selection strategy for k-nearest neighbor construction, while CAN [4] adopts an adaptive neighbor strategy with local connectivity guarantees. Given data correlation, FGNSC [5] aspires to find its good neighbors for each sample, while CLR [6] and SGL [7] attempt to learn a structured graph through respective Laplacian rank constraints.

As a prominent representative of graph learning methods, Sparse Subspace Clustering (SSC) [8], [9] must be mentioned. Thanks to its self-expressiveness property, SSC has achieved gratifying performance, but there is still room for improvement in many aspects. Some works [10], [11] explore SSC in the absence of entries, and others [12], [13] analyze the impact of noise on SSC. Some efforts [14], [15] are also spent on scalability, while most works focus more on the sparsity [16], [17] and low-rank nature [18], [19] of SSC. In addition, some works incorporate transform learning [20], [21], [22], [23] or deep learning [24], [25]. As can be seen, there is still insufficient attention on memory requirements and operational efficiency.

In recent years, people are no longer satisfied with single-view learning, and then turn to multi-view learning. This is because single-view information sometimes cannot provide enough evidence to identify categories, while multi-view information can make up for this deficiency. Given sample features from multiple perspectives, MLAN [26] aims at the consensus graph learning. Given data correlations from multiple perspectives, AMGL [27] aims to learn consistent spectral embeddings, while SwMC [28] focuses on multi-graph fusion. Given spectral embeddings from multiple perspectives, AWP [29] refers to finding the same cluster indicator through spectral rotation. GMC [30] performs multi-graph fusion while learning a graph from each view. MCGC [31] learns the spectral embeddings of each view while exploring a consistent graph. CDMGC [32] uses the original graphs from each perspective to learn and fuse new ones. As an important branch of graph learning, multi-view extensions of SSC have also been greatly developed, such as GLMSC [33], MCLES [34] and FingerMVC [35].

In multi-view scenarios, computational costs and memory consumption become even more prominent. Therefore, many anchor-based multi-view clustering algorithms have been pro-

posed to enhance model scalability. For example, LMVSC [36] learns an anchor graph for each view and then concatenates them. As an enhancement of LMVSC, FDAGF [37] learns multiple anchor graphs for each view. Besides, AEVC [38] introduces an anchor enhancement strategy to capture view correlations. EMVGC-LG [39] preserves both the global and local anchor graph structures. CAMVC [40] learns anchors with cluster structures. CGMAA [41] and FMVACC [42] addresses the issue of anchor alignment.

However, the aforementioned anchor-based algorithms still face some limitations. Their computational complexity is linear with respect to the number of samples n , but quadratic or even cubic with respect to the number of anchors m . Clearly, increasing m leads to a more accurate representation of the correlations between samples, but also incurs a significant time cost. In this paper, we further focus on model efficiency and scalability, thus proposing Simplified Scalable Subspace Clustering (3SC). Particularly, 3SC when $m = n$ is called Simplified Subspace Clustering (2SC). By imposing constraints and performing algebraic operations, the original scalable subspace clustering is approximately transformed into 3SC. Furthermore, thanks to this simple form, a separate fusion strategy is designed, making 3SC and 2SC suitable for multi-view data, with remarkable performance in both effectiveness and efficiency. We refer to these two multi-view models as Multi-View Simplified Scalable Subspace Clustering (MV3SC) and Multi-View Simplified Subspace Clustering (MV2SC), respectively. Overall, the main contributions are summarized as follows.

- Compared with SSC, our 2SC is simpler in form, allowing for more friendly optimization. Therefore, 2SC is a powerful alternative to SSC.
- Compared with 2SC, our 3SC is good at balancing effectiveness and efficiency, with higher scalability. Further, an optimization applicable to both 2SC and 3SC has been designed, which performs well in sparsity control.
- Thanks to our separate fusion strategy, MV3SC and MV2SC can fully utilize multi-view information. They not only avoid alternate optimization, but also ensure the convexity of the two subproblems.

II. RELATED WORKS

This section is described from three parts: anchor selection, single-view graph learning and multi-view subspace clustering.

A. Anchor Selection

Standard subspace clustering employs the full set of samples as a dictionary to learn the correlations between any pair of samples. However, as the number of samples grows, the resulting time cost and memory demand often become unacceptable. To address this, scalable subspace clustering has been proposed, which only explores the correlations between all samples and a set of representative samples. These representative samples are often referred to as anchors, and their selection becomes a critical step. Random selection is the most primitive and straightforward strategy, but it is not popular because the selected samples lack sufficient representativeness.

Commonly used strategies include k-means, Balanced K-means based Hierarchical K-means (BKHK) [43], and feature scoring [44], [45]. K-means can generate m cluster centroids, and the m samples closest to these centroids are used as anchors. BKHK recursively bisects the samples, generating 2^l centroids, where l is the number of bisections. Feature scoring iteratively searches for anchors, recalculating the variance between features of unselected samples in each iteration and selecting the sample with the highest variance as the anchor.

B. Single-View Graph Learning

The spectral clustering performance depends not only on the algorithm itself, but also on the learned graph, which has led to a series of studies on graph learning.

Generally, the similarity between samples is inversely proportional to their distance. Based on this idea, CAN [4] further imposes a rank constraint to force the learned graph to have a clearer structure, and its objective function is

$$\min_{\Theta} \sum_{i=1}^n \sum_{j=1}^n \left(\|x_i - x_j\|_2^2 \theta_{ij} + \gamma \theta_{ij}^2 \right) \\ \text{s.t. } \theta_i^T \mathbf{1} = 1, \mathbf{0} \leq \theta_i \leq \mathbf{1}, \text{rank}(\mathbf{L}_{\Theta}) = n - c \quad (1)$$

where θ_{ij} indicates the similarity between the i -th sample x_i and the j -th sample x_j . $\mathbf{L}_{\Theta}$ represents the Laplacian matrix of Θ . γ is used to control the graph sparsity. Directly dealing with the rank constraint is difficult. Hence, Eq. (1) can be further relaxed as follows.

$$\min_{\Theta} \sum_{i=1}^n \sum_{j=1}^n \left(\|x_i - x_j\|_2^2 \theta_{ij} + \gamma \theta_{ij}^2 \right) + 2\lambda \text{Tr}(\mathbf{F}^T \mathbf{L}_{\Theta} \mathbf{F}) \\ \text{s.t. } \theta_i^T \mathbf{1} = 1, \mathbf{0} \leq \theta_i \leq \mathbf{1}, \mathbf{F} \in \mathbb{R}^{n \times c}, \mathbf{F}^T \mathbf{F} = \mathbf{I} \quad (2)$$

When λ is large enough, this method approaches spectral clustering, forcing the first c smallest eigenvalues are 0.

SSC [8], [9] takes another idea: each sample can be represented by a linear combination of other samples, as shown below.

$$\min_{\Theta \in \Omega} \|X - X\Theta\|_F^2 + \Phi(\Theta) \quad (3)$$

The linear combination matrix Θ can be viewed as a graph. Ω represents a set of certain constraints on Θ . Two specific examples of Eq. (3) are as follows.

$$\min_{\Theta} \|X - X\Theta\|_F^2 + \gamma \|\Theta\|_1 \\ \text{s.t. } \text{diag}(\Theta) = \mathbf{0} \quad (4)$$

$$\min_{\Theta} \|X - X\Theta\|_F^2 + \gamma \|\Theta\|_F^2 \\ \text{s.t. } \Theta \geq \mathbf{0}, \Theta^T \mathbf{1} = \mathbf{1} \quad (5)$$

C. Multi-View Subspace Clustering

Strategies for multi-view subspace clustering are illustrated through specific examples below. The objective function of FPMVS-CAG [46] is as follows:

$$\min_{\mathbf{W}^{(v)}, \mathbf{A}, \mathbf{Z}, \alpha} \sum_{v=1}^{n_v} \alpha_v^2 \|\mathbf{X}^{(v)} - \mathbf{W}^{(v)} \mathbf{A} \mathbf{Z}\|_F^2 \\ \text{s.t. } \mathbf{W}^{(v)T} \mathbf{W}^{(v)} = \mathbf{I}, \mathbf{A}^T \mathbf{A} = \mathbf{I}, \alpha^T \mathbf{1} = 1,$$

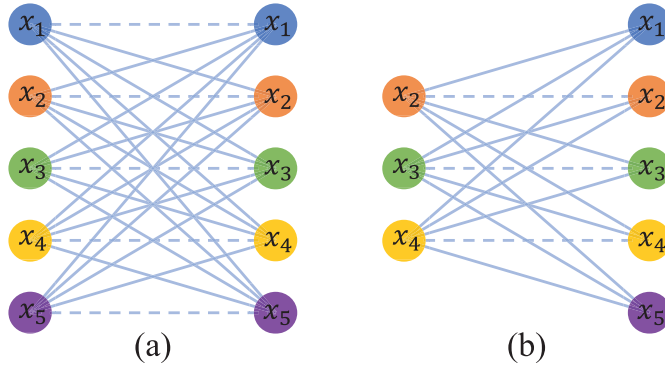


Fig. 1. Standard sparse subspace clustering employs complete samples as a dictionary, described in (a). Scalable sparse subspace clustering is depicted in (b), with representative samples as a dictionary.

$$\alpha \geq 0, Z^T \mathbf{1} = \mathbf{1}, Z \geq 0 \quad (6)$$

It projects anchors from different views into the same latent space through $W^{(v)}$, sharing the same anchor graph Z . The objective function of SMVAGC-SF [47] is as follows:

$$\begin{aligned} \min_{A^{(v)}, Z^{(v)}, R^{(v)}, H, \alpha} & \sum_{v=1}^{n_v} \alpha_v^2 \|X^{(v)} - A^{(v)} Z^{(v)}\|_F^2 \\ & + \beta \|R^{(v)} Z^{(v)} - H\|_F^2 \\ \text{s.t.} & A^{(v)T} A^{(v)} = I, R^{(v)T} R^{(v)} = I, H^T H = I, \\ & \alpha^T \mathbf{1} = 1, \alpha \geq 0, Z^{(v)T} \mathbf{1} = \mathbf{1}, Z^{(v)} \geq 0 \end{aligned} \quad (7)$$

It learns an anchor graph $Z^{(v)}$ for each view and projects these anchor graphs into the same latent space H . The objective function of SFMC [44] is as follows:

$$\begin{aligned} \min_{Z^{(v)}, \alpha} & \left\| \sum_{v=1}^{n_v} \alpha_v Z^{(v)} - Z \right\|_F^2 \\ \text{s.t.} & \alpha^T \mathbf{1} = 1, \alpha \geq 0, Z^T \mathbf{1} = \mathbf{1}, Z \geq 0, \\ & \text{rank}(L_S) = n + m - c \end{aligned} \quad (8)$$

It constructs an anchor graph $Z^{(v)}$ for each view and then fuses these anchor graphs into a unified anchor graph Z . Unexplained symbols and constraints are not the focus of our discussion; please refer to the original papers for details.

III. SIMPLIFIED SPARSE SUBSPACE CLUSTERING

This section is described from six parts: motivation, simplified formulation, optimization, parameter determination, convergence and complexity.

A. Motivation

In sparse subspace clustering, self-expression learning is well known and adopted. As shown in Figure 1(a), each sample is represented by other samples, and each sample also represents other samples. In other words, it uses the feature combination of other samples to reconstruct the features of each sample itself, i.e.

$$X = X\Theta \quad (9)$$

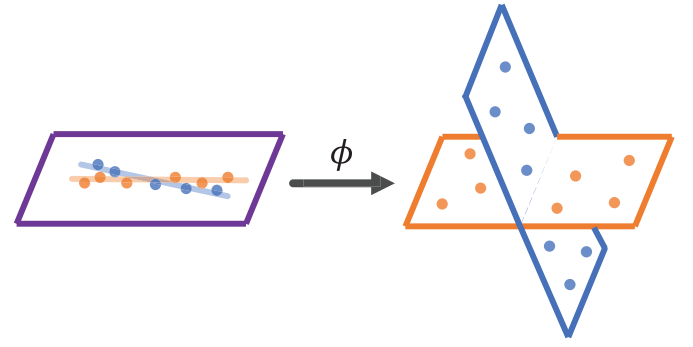


Fig. 2. The original subspaces are mapped to a higher dimensional space for better separation.

where $X \in \mathbb{R}^{d \times n}$ is a complete data matrix consisting of n d -dimensional samples, and $\Theta \in \mathbb{R}^{n \times n}$ implies the association between any sample pairs.

By sampling representative samples as dictionaries instead of all samples, the scalability of sparse subspace clustering will be further improved. These representative samples might as well be called anchors. Naturally, scalable sparse subspace clustering is depicted in Figure 1(b) and has the following concrete form.

$$X = AZ \quad (10)$$

where $A \in \mathbb{R}^{d \times m}$ is a representative data matrix composed of m d -dimensional samples, and $Z \in \mathbb{R}^{m \times n}$ implies the association between anchors and samples. In order to get Z , there is usually the following objective function.

$$\begin{aligned} \min_Z & \|X - AZ\|_F^2 + \gamma \|Z\|_F^2 \\ \text{s.t.} & Z \geq 0, Z^T \mathbf{1} = \mathbf{1} \end{aligned} \quad (11)$$

Furthermore, this main problem can be decomposed into n sub-problems by column, as shown below.

$$\begin{aligned} \min_{z_j} & z_j^T (A^T A + \gamma I) z_j + x_j^T A z_j \\ \text{s.t.} & z_j \geq 0, z_j^T \mathbf{1} = 1 \end{aligned} \quad (12)$$

Obviously, each subproblem is a standard quadratic programming problem with m -dimensional variables, requiring a computational complexity of $O(m^3)$. Therefore, the complete solution requires a computational complexity of $O(nm^3)$. As the number of anchors increases, the solution efficiency will drop significantly. In addition, it also faces the problem of difficulty in controlling sparsity and balance. In the next subsection, we will simplify scalable subspace clustering, thereby constructing an efficient, sparsity-controllable, and distribution-balanced objective function.

B. Simplified Formulation

It is generally believed that samples located in the same subspace can reconstruct each other. However, the original subspaces may not be well separated from each other, as shown in Figure 2. A feasible solution is to map these original subspaces to a higher dimensional space, i.e.

$$\phi(X) = \phi(A)Z \quad (13)$$

where $\phi(\cdot)$ is a mapping from the current space to the reproducing kernel Hilbert space. However, instead of directly solving the above equation, we will leverage its properties to further explore a more convenient objective function.

Decompose $\phi(\mathbf{X}) = \phi(\mathbf{A})\mathbf{Z}$ column-wise, with each sample represented by the anchors as

$$\phi(\mathbf{x}_j) = \sum_{\bar{i} \neq j} z_{ij} \phi(\mathbf{a}_i) \quad (14)$$

The i -th anchor $\mathbf{a}_i$ corresponds to the $\bar{i}$ -th sample $\mathbf{x}_{\bar{i}}$. Multiply both sides by $\phi(\mathbf{x}_j)$, the following equation is established.

$$\phi(\mathbf{x}_j)^T \phi(\mathbf{x}_j) - \sum_{\bar{i} \neq j} z_{ij} \phi(\mathbf{a}_i)^T \phi(\mathbf{x}_j) = 0 \quad (15)$$

By imposing the constraint $\sum_{i=1}^m z_{ij} = 1$, the following relationship is further obtained.

$$\left(\sum_{i=1}^m z_{ij} \right) \phi(\mathbf{x}_j)^T \phi(\mathbf{x}_j) - \sum_{\bar{i} \neq j} z_{ij} \phi(\mathbf{a}_i)^T \phi(\mathbf{x}_j) = 0 \quad (16)$$

It is easy to know

$$\sum_{i=1}^m \sum_{j=1}^n (l_j - k_{ij}) z_{ij} + \sum_{i=1}^m k_{i\bar{i}} z_{i\bar{i}} = 0 \quad (17)$$

where $l_j = \phi(\mathbf{x}_j)^T \phi(\mathbf{x}_j)$ and $k_{ij} = \phi(\mathbf{a}_i)^T \phi(\mathbf{x}_j)$ is a kernel mapping between anchors and samples. There is no need to calculate the kernel mapping between all samples. If c_{ij} is pre-calculated as

$$c_{ij} = \begin{cases} l_j - k_{ij} & \text{if } \bar{i} \neq j \\ l_j & \text{otherwise} \end{cases} \quad (18)$$

there will be preliminary formulation for Simplified Scalable Subspace Clustering (3SC) as follows.

$$\min_{z_{ij} \geq 0} \sum_{i=1}^m \sum_{j=1}^n c_{ij} z_{ij} \quad \text{s.t.} \quad \sum_{i=1}^m z_{ij} = 1 \quad (19)$$

Furthermore, a more balanced sample distribution is required, with the following objective function.

$$\min_{z_{ij} \geq 0} \sum_{i=1}^m \sum_{j=1}^n c_{ij} z_{ij} \quad \text{s.t.} \quad \sum_{i=1}^m z_{ij} = 1, \quad \sum_{j=1}^n z_{ij} = \frac{n}{m} \quad (20)$$

Surely, the initial problem has been approximately transformed into a simpler one, with more efficient optimization. In fact, this form potentially reflects a special supply-demand relationship. Concretely, each sample is a supplier holding one unit of goods, but only m samples are demanders requiring $\frac{n}{m}$ unit of goods. Suppliers must supply the goods they hold, while demanders must receive the supplied goods. They cannot keep their own goods. Let c_{ij} and z_{ij} be the transport cost and transport plan from supplier j to demander i , respectively. Our goal is to find a transport plan that minimizes the total cost.

Unlike the strict optimal transport, the unit of goods here is continuous and separable. Additionally, in order to facilitate the solution and control the sparsity of the learned graph, a regularizer $\Phi(z_{ij})$ is added as shown below.

$$\min_{z_{ij} \geq 0} \sum_{i=1}^m \sum_{j=1}^n c_{ij} z_{ij} + \Phi(z_{ij})$$

$$\text{s.t.} \quad \sum_{i=1}^m z_{ij} = 1, \quad \sum_{j=1}^n z_{ij} = \frac{n}{m} \quad (21)$$

Here, $\Phi(z_{ij}) = \gamma z_{ij}^2$ is adopted in our model. As γ decreases, there will be fewer and fewer transport channels for goods, and Eq. (21) is closer to the original optimal transport. When γ approaches 0, this will lead to the goods held by each supplier can only be received by a single demander, which makes the graph too sparse and cannot well express the correlation between samples.

In addition, we can even impose a stricter constraint $\|\mathbf{z}_j\|_0 \leq k$ as shown in Eq. (22), where $\mathbf{z}_j$ represents the j -th column of $\mathbf{Z}$. The number of non-zero elements in each column of $\mathbf{Z}$ does not exceed k .

$$\begin{aligned} \min_{z_{ij} \geq 0} & \sum_{i=1}^m \sum_{j=1}^n c_{ij} z_{ij} + \gamma z_{ij}^2 \\ \text{s.t.} & \sum_{i=1}^m z_{ij} = 1, \quad \sum_{j=1}^n z_{ij} = \frac{n}{m}, \quad \|\mathbf{z}_j\|_0 \leq k \end{aligned} \quad (22)$$

C. Optimization

By introducing Lagrange multipliers μ and ν , there is the following min-max problem.

$$\min_{z_j \in S_z^k} \max_{\mu, \nu} \sum_{j=1}^n \mathbf{z}_j^T \mathbf{c}_j - \mu^T (\mathbf{Z}\mathbf{1} - \mathbf{l}) - \nu^T (\mathbf{Z}^T \mathbf{1} - \mathbf{1}) + \gamma \mathbf{z}_j^T \mathbf{z}_j \quad (23)$$

where $S_z^k = \{\mathbf{z} \in \mathbb{R}^m \mid \|\mathbf{z}\|_0 \leq k, \mathbf{z} \geq \mathbf{0}\}$ and $\mathbf{l} = \frac{n}{m} \mathbf{1}$. Next, by simple mathematical transformations, there is the following max-min problem.

$$\max_{\mu, \nu} \mu^T \mathbf{l} + \nu^T \mathbf{1} + \sum_{j=1}^n \min_{z_j \in S_z^k} \mathbf{z}_j^T (\mathbf{c}_j - \mu - \nu_j \mathbf{1}) + \gamma \mathbf{z}_j^T \mathbf{z}_j \quad (24)$$

Finally, the dual formulation of our primal problem holds.

$$\max_{\mu, \nu} \mu^T \mathbf{l} + \nu^T \mathbf{1} - \sum_{j=1}^n \max_{z_j \in S_z^k} \mathbf{z}_j^T (\mu + \nu_j \mathbf{1} - \mathbf{c}_j) - \gamma \mathbf{z}_j^T \mathbf{z}_j \quad (25)$$

Let $\mathbf{S} = \mu \mathbf{1}^T + \mathbf{1} \nu^T - \mathbf{C}$ and $\mathbf{s}_j$ be column j of $\mathbf{S}$, the conjugate function $\Phi^*(\mathbf{s}_j)$ of our regularizer $\Phi(\mathbf{z}_j)$ is

$$\Phi^*(\mathbf{s}_j) = \max_{z_j \in S_z^k} \mathbf{z}_j^T \mathbf{s}_j - \Phi(\mathbf{z}_j) \quad (26)$$

Further, there are derivatives $\nabla \Phi^*(\mathbf{s}_j)$ as known.

$$\begin{aligned} \nabla \Phi^*(\mathbf{s}_j) &= \arg \max_{z_j \in S_z^k} \mathbf{z}_j^T \mathbf{s}_j - \Phi(\mathbf{z}_j) \\ &= \arg \min_{z_j \in S_z^k} \|\mathbf{z}_j\|_2^2 - \frac{\mathbf{z}_j^T \mathbf{s}_j}{\gamma} + \frac{\|\mathbf{s}_j\|_2^2}{4\gamma^2} \\ &= \arg \min_{z_j \in S_z^k} \left\| \frac{\mathbf{s}_j}{2\gamma} - \mathbf{z}_j \right\|_2^2 = \frac{[\mathbf{s}_j]_+^k}{2\gamma} \end{aligned} \quad (27)$$

When $s_{ij} > 0$ and belongs to the top- k largest elements of $\mathbf{s}_j$, $[s_{ij}]_+^k = s_{ij}$, otherwise $[s_{ij}]_+^k = 0$.

According to the above analysis, our dual problem can be rewritten as follows.

$$\min_{\mu, \nu} \sum_{j=1}^n \Phi^*(s_j) - \mu^T l - \nu^T \mathbf{1} \quad (28)$$

Therefore, its partial derivatives are obtained as

$$\frac{\partial \mathcal{L}}{\partial \mu} = \sum_{j=1}^n \frac{[s_j]_+^k}{2\gamma} - l \quad (29)$$

$$\frac{\partial \mathcal{L}}{\partial \nu} = \sum_{i=1}^m \frac{[\tilde{s}_i]_+^k}{2\gamma} - \mathbf{1} \quad (30)$$

where $\tilde{s}_i$ is the i -th column of S^T . Obviously, the optimal solution can be found by any gradient-based algorithm, such as L-BFGS [48]. When reaching the extreme points μ^* and ν^* , the optimal transport plan Z^* can be recovered in the following way.

$$z_j^* = \nabla \Phi^*(\mu^* + \nu_j^* \mathbf{1} - c_j) \quad (31)$$

Algorithm 1 3SC Solver

Input: Dataset X , anchor number m , neighbor number k and cluster number c .

Output: Learned anchor graph Z^* .

- 1: Select anchors by Subsection II-A.
 - 2: Generate a cost matrix C by Eq. (18).
 - 3: Initialize $\mu = \mathbf{0}$ and $\nu = \mathbf{0}$.
 - 4: **while** non-convergence **do**
 - 5: Compute the gradient of μ by Eq. (29).
 - 6: Compute the gradient of ν by Eq. (30).
 - 7: Update μ and ν by any gradient-based algorithm, such as L-BFGS.
 - 8: **end while**
 - 9: Obtain Z^* by Eq. (31).
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For the sake of intuition, our methodological procedure is summarized in Algorithm 1.

D. Parameter Determination

Although our objective function has been greatly simplified, there are still some difficulties in setting the parameter γ . This subsection further explores the parameter setting. Remove the constraints $\sum_{j=1}^n z_{ij} = \frac{n}{m}$ and $\|z_j\|_0 \leq k$ from Eq. (22), the following function exists.

$$\begin{aligned} \min_{z_{ij}} \quad & \sum_{i=1}^m \sum_{j=1}^n c_{ij} z_{ij} + \gamma z_{ij}^2 \\ \text{s.t.} \quad & \sum_{i=1}^m z_{ij} = 1, \quad z_{ij} \geq 0 \end{aligned} \quad (32)$$

Decompose Eq. (32) into n sub-problems by column, and the j -th sub-problem is as follows.

$$\begin{aligned} \min_{z_j} \quad & \left\| z_j - \frac{\gamma_j - c_j}{2\gamma_j} \right\|_2^2 \\ \text{s.t.} \quad & z_j^T \mathbf{1} = 1, \quad z_j \geq \mathbf{0} \end{aligned} \quad (33)$$

According to Karush-Kuhn-Tucker conditions, we know

$$z_{ij} = \left[\eta_j + \frac{\gamma_j - c_{ij}}{2\gamma_j} \right]_+ \quad (34)$$

where η_j is a Lagrangian multiplier. If the j -th column has at most k non-zero elements, then the following equality and inequality hold.

$$\sum_{i=1}^k \left(\eta_j + \frac{\gamma_j - \tilde{c}_{ij}}{2\gamma_j} \right) = 1 \quad (35)$$

$$\eta_j + \frac{\gamma_j - \tilde{c}_{k+1,j}}{2\gamma_j} \leq 0 \quad (36)$$

where $\tilde{c}_j$ is the ascending rearrangement of c_j , i.e. $c_{1j} \leq c_{2j} \leq \dots \leq c_{kj}$. Furthermore, the following inequality must be guaranteed.

$$\gamma_j \leq \frac{k\tilde{c}_{k+1,j} - \sum_{i=1}^k \tilde{c}_{ij}}{2} \quad (37)$$

Therefore, γ can be determined indirectly by the number of neighbors k as follows.

$$\gamma = \frac{1}{2n} \sum_{j=1}^n \left(k\tilde{c}_{k+1,j} - \sum_{i=1}^k \tilde{c}_{ij} \right) \quad (38)$$

Usually, sparse graphs are expected, so k is a small positive integer with a smaller range of values than γ .

E. Convergence

For any function $f(a, b)$, the following inequality holds.

$$\min_a \max_b f(a, b) \geq \max_b \min_a f(a, b) \quad (39)$$

Therefore, Problem (25) is the lower bound of Problem (23) (weak duality). Further, if $k \geq n$, then Problem (22) degenerates into Problem (21) and satisfies the Slater's condition. In this case, Problem (25) and Problem (23) are equivalent (strong duality).

For Problem (25), $\Phi(z_j)$ is potentially non-convex on $\mathcal{S}_z^k$. Fortunately, even if the original function is not convex, its conjugate function $\Phi^*(s_j)$ must be convex. Moreover, the gradient $\nabla \Phi^*(s_j)$ can be accurately calculated. Therefore, Problem (25) is actually a concave maximization problem about s_j . Through any gradient-based algorithm, this problem can converge to the global optimal solution.

F. Complexity

1) *Time Complexity:* In each iteration, the computational complexity of s_j is $O(m)$. If $k < m$, the top- k lookup needs to be executed with $O(m \ln k)$ computational complexity. Next, some element-wise operations require the computational complexity of $O(m)$. Since there are n such column vectors, the overall need is $O(nm)$. Finally, the sum of n vectors also requires $O(nm)$. Therefore, a single gradient calculation requires $O(nm)$. Once the gradient is given, L-BFGS only needs $O(n+m)$ to update μ and ν . In total, our computational complexity for a single iteration is $O(nm)$. If $m \ll n$, linear complexity can be approximated. Since our algorithm basically only involves the element-wise operation, the underlying parallelization and other strategies make it actually faster.

2) *Space Complexity*: The transport cost C , auxiliary variable S , Lagrange multipliers μ and ν require storage complexities of $O(mn)$, $O(mn)$, $O(m)$ and $O(n)$, respectively. Therefore, the overall space complexity is $O(nm)$. When $m \ll n$, linear complexity can be further achieved.

IV. EXTENSION TO MULTI-VIEW CLUSTERING

This section is described from five parts: motivation, separate fusion strategy, optimization, convergence and complexity.

A. Motivation

Given multi-view data $\{X^{(v)}\}$, with corresponding anchors $\{A^{(v)}\}$. Let $X = [\alpha_1 X^{(1)}; \dots; \alpha_{n_v} X^{(n_v)}]$ and $A = [\alpha_1 A^{(1)}; \dots; \alpha_{n_v} A^{(n_v)}]$, Eq. (11) is extended as follows:

$$\begin{aligned} \min_{Z, \alpha} \sum_{v=1}^{n_v} \alpha_v^2 \|X^{(v)} - A^{(v)} Z\|_F^2 + \gamma \|Z\|_F^2 \\ \text{s.t. } \alpha^T \mathbf{1} = 1, \alpha \geq \mathbf{0}, Z^T \mathbf{1} = \mathbf{1}, Z \geq \mathbf{0} \end{aligned} \quad (40)$$

However, Eq. (40) not only fails to escape the issues mentioned in Subsection III-A, but also faces the following new problems. Z and α need to be solved alternately, and the two subproblems themselves require iteration, which increases the computational burden. In addition, the smaller the view-specific loss, the greater the view-specific weight, while the loss cannot necessarily reflect the view-specific clustering ability. In the next subsection, we will adopt a separate fusion strategy for multi-view extensions, better measuring the inter-view difference and avoid alternate optimization, so as to achieve more robust and efficient multi-view clustering.

B. Separate Fusion Strategy

Because the construction of cost matrix completely relies on the kernel matrix, the difference analysis can focus on the kernel matrix. Let $K^{(v)}$ have the following economical singular value decomposition.

$$K^{(v)} = U_c^{(v)} \Sigma_c^{(v)} V_c^{(v)T} \quad (41)$$

where $\Sigma_c^{(v)}$ is a diagonal matrix composed of the top c singular values, $U_c^{(v)}$ and $V_c^{(v)}$ are the corresponding left and right singular vector matrices, respectively. Mark the desirable kernel as $K^* = \sum_{v=1}^{n_v} \alpha_v K^{(v)}$. The ideal relationship between K^* and each $K^{(v)}$ is as follows.

$$U_c^{(v)T} K^* = U_c^{(v)T} K^{(v)} = \Sigma_c^{(v)} V_c^{(v)T} \quad (42)$$

Hence, our weight can be determined by

$$\begin{aligned} \min_{\alpha} \sum_{v=1}^{n_v} \|U_c^{(v)T} K^* - \Sigma_c^{(v)} V_c^{(v)T}\|_F^2 \\ \text{s.t. } K^* = \sum_{v=1}^{n_v} \alpha_v K^{(v)}, \alpha^T \mathbf{1} = 1, \alpha \geq \mathbf{0} \end{aligned} \quad (43)$$

Specifically, K^* is required to minimize the difference with each $K^{(v)}$ from different views in the projection space. Once

the extreme point is reached, the consensus cost matrix can be constructed by

$$c_{ij}^* = \begin{cases} \sum_{v=1}^{n_v} \alpha_v l_j^{(v)} - k_{ij}^* & \text{if } \bar{i} \neq j \\ \sum_{v=1}^{n_v} \alpha_v l_j^{(v)} & \text{otherwise} \end{cases} \quad (44)$$

Further, 3SC Solver can be invoked. Obviously, the whole multi-view clustering process is decoupled into two steps, avoiding alternating iterations. Relying on 3SC, the multi-view extension is friendly to operation efficiency, sparsity control, and balanced distribution.

Algorithm 2 MV3SC Solver

Input: Multi-view dataset $\{X^{(v)}\}_{v=1}^{n_v}$, anchor number m , neighbor number k and cluster number c .

Output: Weight α^* and learned anchor graph Z^* .

- 1: Select anchors by Subsection II-A.
 - 2: Generate $l^{(v)}$ and $K^{(v)}$ for each view.
 - 3: Calculate left singular vector matrix $U_c^{(v)}$, right singular vector matrix $V_c^{(v)}$ and singular value matrix $\Sigma_c^{(v)}$ of $K^{(v)}$ for each view.
 - 4: Initialize $\alpha = \mathbf{1}/n_v$, $\mu = \mathbf{0}$ and $\nu = \mathbf{0}$.
 - 5: Calculate α^* by solving the Problem (47).
 - 6: Generate a consensus cost matrix C^* by Eq. (44).
 - 7: **while** non-convergence **do**
 - 8: Compute the gradient of μ by Eq. (29).
 - 9: Compute the gradient of ν by Eq. (30).
 - 10: Update μ and ν by any gradient-based algorithm, such as L-BFGS.
 - 11: **end while**
 - 12: Obtain Z^* by Eq. (31).
-

C. Optimization

Let $H^{(v)}$ and $f^{(v)}$ satisfy the following equalities.

$$H_{ij}^{(v)} = \text{Tr}(K^{(i)T} U_c^{(v)} U_c^{(v)T} K^{(j)}) \quad (45)$$

$$f_i^{(v)} = \text{Tr}(K^{(i)T} V_c^{(v)} \Sigma_c^{(v)T} U_c^{(v)T}) \quad (46)$$

Eq. (43) can be further organized into

$$\begin{aligned} \min_{\alpha} \alpha^T \left(\sum_{v=1}^{n_v} H^{(v)} \right) \alpha - 2\alpha^T \left(\sum_{v=1}^{n_v} f^{(v)} \right) \\ \text{s.t. } \alpha^T \mathbf{1} = 1, \alpha \geq \mathbf{0} \end{aligned} \quad (47)$$

Obviously, Eq. (47) is an exactly well-defined quadratic programming problem, which can be solved by many existing algorithms, such as interior point method. The former analysis is summarized in Algorithm 2.

D. Convergence

Problem (47) is a standard and convex quadratic programming problem, because of its positive semidefinite Hessian matrix. If at least one vector satisfies the constraint and has a lower bound in the feasible region, the convex quadratic programming problem has a global minimum. The solution and convergence of quadratic programming problem can be referred to [49], [50].

E. Complexity

1) *Time Complexity*: The computational complexity of constructing and solving Eq. (47) are $O(n_v^3 nmc)$ and $O(n_v^3)$, where n , c and n_v are the numbers of samples, clusters and views. The computational complexity of solving Eq. (22) is $O(nm)$. Therefore, the overall complexity is $O(n_v^3 nmc + n_v^3 + nm + nmc)$. Generally, since $n_v \ll n$ and $c \ll n$, the overall complexity is approximately equal to $O(nm)$. If $m \ll n$, linear complexity can be further implemented.

2) *Space Complexity*: The sets $\{\mathbf{K}^{(v)}\}$, $\{\mathbf{U}_c^{(v)}\}$, $\{\mathbf{\Sigma}_c^{(v)}\}$, and $\{\mathbf{V}_c^{(v)}\}$ require storage complexities of $O(nmn_v)$, $O(mcn_v)$, $O(c^2 n_v)$, and $O(ncn_v)$, respectively. Additionally, $\{\mathbf{H}^{(v)}\}$ and $\{\mathbf{f}^{(v)}\}$ require storage complexities of $O(n_v^3)$ and $O(n_v^2)$, respectively. In general, since $n_v \ll n$ and $c \ll n$, the overall space complexity is approximately $O(nm)$. If $m \ll n$, linear complexity can be further achieved.

V. EXPERIMENT

This section is divided into four aspects: experimental details, single-view clustering, multi-view clustering and multi-kernel clustering.

A. Experimental Details

The experimental details are elaborated in four parts: datasets, baselines, settings and metrics.

1) *Datasets*: In single-view experiments, 6 classic datasets are used for comparison, including COIL20,¹ ORL², ISO-LET³, LUNG⁴, MSRA and USPS.⁵ In addition, 3 large datasets, STL10,⁶ MNIST⁷ and FMNIST,⁸ are employed specifically for the comparison of near-linear single-view algorithms. In multi-view and multi-kernel experiments, another 6 classic datasets are adopted for comparison, including MSRC,⁹ ALOI,¹⁰ 100Leaves,¹¹ 3Sources¹², BBCNews¹³ and BBCSport fn:uci. In addition, 3 large datasets, Cifar10,¹⁴ Cifar100 fn:uci and YTF10,¹⁵ are employed specifically for the comparison of near-linear multi-view algorithms. Their statistics are described in Table I.

2) *Baselines*: In single-view experiments, our 3SC and 2SC are compared with the following seven algorithms: KMM [51], LCSOG [52], FCAG [53], CAN [4], CLR [6], SSC [9] and SGL [7]. The first three have near-linear time complexity, while that of the last four is non-linear. In multi-view experiments, our MV3SC and MV2SC are compared with the following thirteen algorithms: LMVSC [36], SFMC

TABLE I
DESCRIPTION FOR SINGLE-VIEW AND MULTI-VIEW DATASETS

Dataset	#Views	#Instances	#Classes	#Features
COIL20	1	1440	20	1024
ORL	1	400	40	1024
MSRA	1	1799	12	256
LUNG	1	203	5	3312
ISOLET	1	1560	26	617
USPS	1	9298	10	256
STL10	1	13000	10	2048
MNIST	1	70000	10	784
FMNIST	1	70000	10	784
MSRC	5	210	7	24, 576, 512, 256, 254
ALOI	4	1080	10	64, 64, 77, 13
100Leaves	3	1600	100	64, 64, 64
3Sources	3	169	6	3560, 3631, 3068
BBCNews	4	685	5	4659, 4633, 4665, 4680
BBCSport	2	544	5	3183, 3203
Cifar10	3	50000	10	512, 2048, 1024
Cifar100	3	50000	100	512, 2048, 1024
YTF10	4	38654	10	944, 576, 512, 640

TABLE II
COMMON KERNEL FUNCTIONS

Type	Formulation	Default
Linear	$w\mathbf{a}_i^T \mathbf{x}_j + b$	$w = 1, b = 1$
Gaussian	$\exp\left(-\frac{1}{2\sigma^2} \ \mathbf{a}_i - \mathbf{x}_j\ _2^2\right)$	$\sigma = \sqrt{\frac{1}{nm} \sum_{ij} \ \mathbf{a}_i - \mathbf{x}_j\ _2^2}$
Cauchy	$\left(\frac{1}{\sigma^2} \ \mathbf{a}_i - \mathbf{x}_j\ _2^2 + 1\right)^{-1}$	$\sigma = \sqrt{\frac{1}{nm} \sum_{ij} \ \mathbf{a}_i - \mathbf{x}_j\ _2^2}$
Polynomial	$(w\mathbf{a}_i^T \mathbf{x}_j + b)^p$	$w = 1, b = 0, p = 2$
Sigmoid	$\tanh(w\mathbf{a}_i^T \mathbf{x}_j + b)$	$w = 1, b = 0$

[44], TBGL [45], OMSC [54], CAMVC [40], SMVAGC [47], AMGL [27], SwMC [28], MLAN [26], GMC [30], MCGC [31], CDMGC [32] and MCLES [34]. The first six have near-linear time complexity, while that of the last seven is non-linear. In multi-kernel experiments, our MV3SC and MV2SC are also compared with the following seven algorithms: MKKM [55], LMKKM [55], ONKC [56], HMKC [57], DPMKKM [58], SMKKM [59] and LSMKKM [60]. These classic or state-of-the-art algorithms have a significant correlation with ours.

3) *Settings*: In single-view, multi-view and multi-kernel experiments, our models selectively employ Gaussian or linear kernel on each dataset. Several common kernel functions are summarized in Table II. The parameter γ is determined by the desired number of neighbors k , and k is set by grid $\{1, 2, \dots, 16\}$. Considering the initialization sensitivity and randomness during running, we performed all algorithms 20 times. For graphs with clear connected components, the clustering results are directly obtained. The parameters of all compared algorithms are set by grid search according to the recommendations of their original papers.

4) *Metrics*: In single-view, multi-view and multi-kernel experiments, the performance is evaluated by Accuracy (ACC), Normalized Mutual Information (NMI) and Adjusted

¹<http://www.cs.columbia.edu/CAVE/software/softlib/coil-20.php>

²<http://www.cl.cam.ac.uk/research/dtg/attarchive/facedatabase.html>

³<https://archive.ics.uci.edu/dataset/49/isolet>

⁴<https://archive.ics.uci.edu/dataset/84/lung+cancer>

⁵<https://www.kaggle.com/datasets/bistaumanga/usps-dataset>

⁶<https://cs.stanford.edu/acoates/stl10/>

⁷<http://yann.lecun.com/exdb/mnist/>

⁸<https://github.com/zalandoresearch/fashion-mnist>

⁹<http://yacvid.hayko.at/task.php?did=35>

¹⁰<https://aloi.science.uva.nl/>

¹¹<https://archive.ics.uci.edu/ml/datasets.php>

¹²<http://mlg.ucd.ie/datasets/3sources.html>

¹³<http://mlg.ucd.ie/datasets/segment.html>

¹⁴<https://www.cs.toronto.edu/kriz/cifar.html>

¹⁵<https://www.cs.tau.ac.il/~wolf/ytfaces>

TABLE III

ACCURACY (ACC%±STD%), NORMALIZED MUTUAL INFORMATION (NMI%±STD%), ADJUSTED RAND INDEX (ARI%±STD%) AND RUNTIME (S) FOR SINGLE-VIEW CLUSTERING ALGORITHM COMPARISON

Metric	ACC	NMI	ARI	Time	ACC	NMI	ARI	Time	ACC	NMI	ARI	Time
Dataset	COIL20				ORL				ISOLET			
KMM	74.1±4.9	88.2±1.7	72.3±3.2	6.6E+00	50.3±3.6	67.3±2.5	18.4±3.3	7.6E-01	58.5±0.8	76.7±1.1	50.5±1.7	6.7E+00
LCSOG	73.3±0.0	82.7±0.0	61.5±0.0	4.4E-01	48.8±0.0	69.0±0.0	23.8±0.0	7.0E-02	61.1±0.0	75.8±0.0	53.9±0.0	4.3E-01
FCAG	66.0±2.9	78.7±1.0	61.2±2.0	9.6E-01	59.6±2.1	76.8±0.9	44.1±2.0	1.2E-01	61.2±2.0	75.6±0.9	56.3±2.3	6.6E-01
3SC	82.0±1.0	87.0±0.6	74.8±1.9	1.9E-01	62.2±2.5	76.4±0.6	46.4±1.4	1.5E-01	67.1±1.5	76.9±0.8	57.7±1.3	2.7E-01
CAN	85.8±0.0	92.2±0.0	82.6±0.0	2.0E+00	56.0±0.0	74.6±0.0	32.4±0.0	1.1E-01	60.8±0.0	77.8±0.0	56.0±0.0	2.1E+00
CLR	86.0±0.0	92.8±0.0	83.2±0.0	6.5E+00	63.3±0.0	79.9±0.0	45.7±0.0	3.1E-01	61.4±0.0	78.7±0.0	54.5±0.0	5.2E+00
SSC	86.9±1.0	92.1±0.5	82.4±1.0	5.8E+01	66.4±1.2	81.7±0.4	52.7±1.7	8.4E+00	64.3±1.8	76.2±0.4	57.4±0.8	4.1E+01
SGL	83.5±0.0	91.5±0.0	80.6±0.0	8.5E+00	67.4±1.7	81.7±0.8	54.5±1.5	2.9E-01	60.1±1.6	77.2±0.5	55.0±0.9	1.1E+01
2SC	98.2±0.0	98.2±0.0	96.5±0.0	4.5E-01	70.1±1.2	84.3±0.7	59.7±1.5	1.8E-01	65.4±1.1	79.8±0.2	59.5±0.2	4.6E-01
Dataset	LUNG				MSRA				USPS			
KMM	84.3±2.2	46.1±7.6	53.7±9.8	1.7E-01	51.3±3.5	61.1±3.1	32.6±4.5	5.3E+00	74.1±2.8	75.7±2.0	65.0±3.5	3.6E+02
LCSOG	84.7±0.0	50.4±0.0	59.3±0.0	3.0E-02	53.4±0.0	65.6±0.0	37.6±0.0	4.9E-01	74.6±0.0	79.9±0.0	74.0±0.0	2.0E+01
FCAG	75.5±9.2	51.9±8.6	48.4±9.9	1.1E-01	50.0±1.7	59.9±2.2	35.8±2.2	4.9E-01	74.5±6.5	80.1±4.7	70.5±6.9	1.0E+01
3SC	84.7±0.0	58.8±0.0	67.4±0.0	1.2E-01	57.3±0.0	66.8±0.1	45.4±0.0	3.4E-01	85.6±0.0	81.2±0.0	81.1±0.0	3.9E+00
CAN	85.7±0.0	65.3±0.0	63.2±0.0	3.0E-02	52.5±0.0	59.7±0.0	29.7±0.0	1.9E+00	84.1±0.0	83.2±0.0	80.7±0.0	1.2E+03
CLR	85.7±0.0	63.4±0.0	62.9±0.0	7.0E-02	53.8±0.0	69.2±0.0	42.4±0.0	6.4E+00	80.1±0.0	86.2±0.0	78.1±0.0	1.3E+03
SSC	83.7±0.0	60.5±0.0	69.3±0.0	2.9E+01	54.6±1.6	67.9±0.7	41.6±1.5	3.8E+01	56.1±0.0	68.7±0.0	50.0±0.0	2.4E+03
SGL	81.8±0.0	60.0±0.0	58.0±0.0	8.0E-02	53.9±0.0	69.2±0.3	43.2±0.7	1.4E+01	81.5±0.0	86.0±0.0	79.2±0.0	2.5E+03
2SC	88.7±0.0	69.2±0.0	77.0±0.0	1.2E-01	60.7±0.0	71.8±1.0	48.7±1.9	6.8E-01	90.5±0.0	84.6±0.0	84.7±0.0	3.1E+01

TABLE IV

ACCURACY (ACC%±STD%), NORMALIZED MUTUAL INFORMATION (NMI%±STD%), ADJUSTED RAND INDEX (ARI%±STD%) AND RUNTIME (S) FOR EXTRA NEAR-LINEAR SINGLE-VIEW CLUSTERING ALGORITHM COMPARISON

Metric	ACC	NMI	ARI	Time	ACC	NMI	ARI	Time	ACC	NMI	ARI	Time
Dataset	STL10				MNIST				FMNIST			
KMM	83.3±0.0	80.4±0.0	70.8±0.0	4.5E+02	OM	OM	OM	OM	OM	OM	OM	OM
LCSOG	83.4±0.0	80.3±0.0	70.6±0.0	2.3E+01	83.3±0.0	87.6±0.0	81.8±0.0	1.9E+03	53.7±0.0	57.9±0.0	41.6±0.0	2.7E+03
FCAG	89.8±7.4	83.0±2.2	78.7±4.9	3.8E+02	83.5±0.1	85.1±0.2	79.9±0.2	5.6E+02	57.2±0.4	60.9±1.2	44.0±1.4	5.7E+02
3SC	90.8±0.0	84.3±0.0	81.1±0.0	1.5E+01	96.2±0.0	91.1±0.0	91.9±0.0	4.6E+02	61.3±0.0	64.4±0.0	50.8±0.0	4.3E+02

OM: Out of Memory.

Rand Index (ARI). Their means and standard deviations are reported in experiments.

B. Single-View Clustering

1) *Performance*: As shown in Table III, 3SC is superior to other algorithms with near-linear or even nonlinear complexity, while 2SC almost achieves the best performance among all compared algorithms. As shown in Table IV 3SC ran successfully on three additional large datasets, achieving optimal performance in all cases, while KMM encountered memory shortages.

2) *Runtime*: Table III shows that, 2SC has a little time overhead, and 3SC further reduces this time overhead. In terms of efficiency, both are not only great improvements to SSC, but also better than other algorithms. Furthermore, Table IV shows that, 3SC remains efficient when running on large datasets.

3) *Convergence*: Figure 3 depicts the convergence curves. As the iterations increase, the losses of 3SC and 2SC first drop sharply and then level off. This implies that our models are able to converge rapidly and stably.

4) *Neighbor Number*: Take 1, 2, ..., 16 neighbor numbers in turn to observe the accuracy change trends of 3SC and

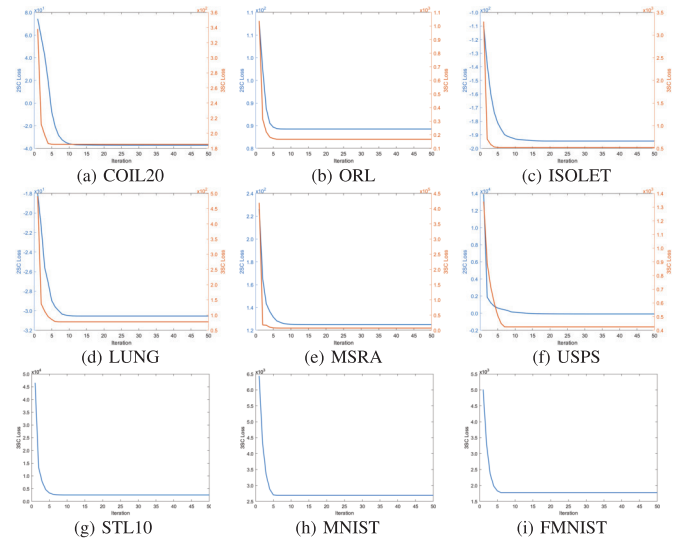


Fig. 3. Convergence curves of 3SC and 2SC during optimization.

2SC with Gaussian/linear kernel (3SC_G/L and 2SC_G/L). As captured in Figure 4, the increase in neighbor numbers

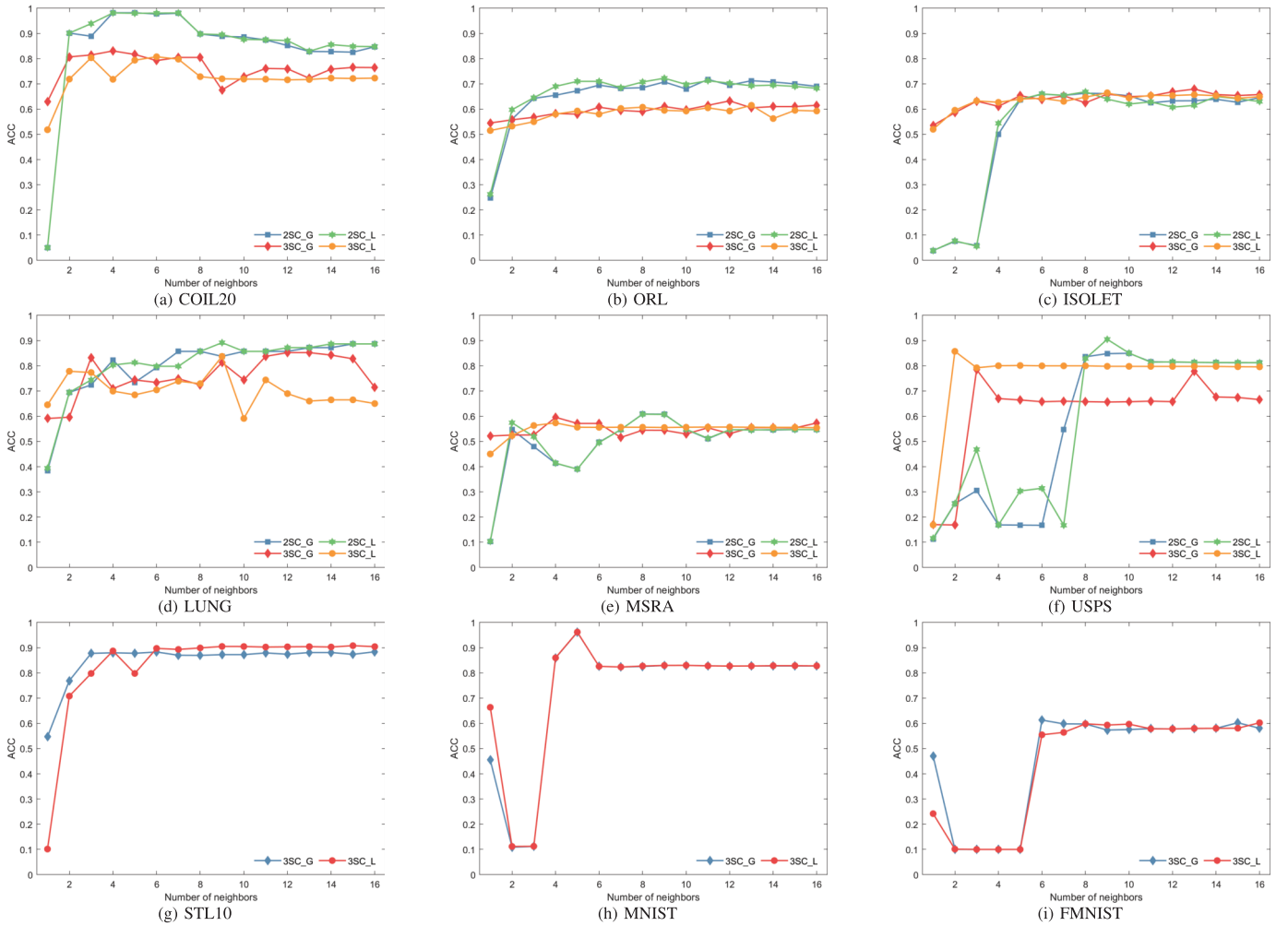


Fig. 4. Accuracy of 2SC and 3SC under different numbers of neighbors.

TABLE V
ACCURACY OF 3SC UNDER DIFFERENT ANCHOR STRATEGIES

Dataset	Random	K-means	VDA
COIL20	78.3	83.2	82.5
ORL	58.2	61.7	62.2
MSRA	57.2	57.2	57.4
LUNG	82.3	85.2	83.7
ISOLET	62.5	64.2	67.6
USPS	77.8	85.6	68.0
STL10	77.5	90.8	87.8
MNIST	95.4	82.2	96.2
FMNIST	56.6	59.5	61.3

leads to an initial increase in accuracy and then a gradual flattening. This implies that our model parameters are easy to set, only requiring a small number of neighbors.

5) *Anchor Ratio*: Take 10%, 15%, ..., 60% samples as anchors in turn to observe the performance change trend of 3SC. As captured in Figure 5, an excessive number of anchors fails to bring significant performance improvement. This implies that our model is lightweight and requires only a few anchors.

6) *Anchor Selection*: Several anchor selection strategies are tested for 3SC, including random sampling, k-means sampling, and VDA sampling. As described in Table V, random sampling is generally weaker than the latter two, while k-means sampling and VDA sampling show mixed performance. When the number of anchors is large, the differences among the three methods may diminish, but VDA is preferable due to its lower computational cost compared to k-means. However, when the number of anchors is small, VDA may fail, whereas k-means remains effective and is therefore recommended.

7) *Kernel Type*: More types of kernels are tested, including Gaussian, Cauchy, linear, polynomial and sigmoid kernels. As reported in Table VI, 3SC with Gaussian kernel generally performs better, while 2SC tends to be equipped with linear kernel. Although the best performance will occasionally be shown on other kernels, the Gaussian and linear kernels are usually not far behind them at this time. Hence, there is no need to spend too much energy on kernel selection.

C. Multi-View Clustering

1) *Performance*: As shown in Table VII, MV3SC is superior to other algorithms with near-linear or even non-linear complexity, while MV2SC almost achieves the best

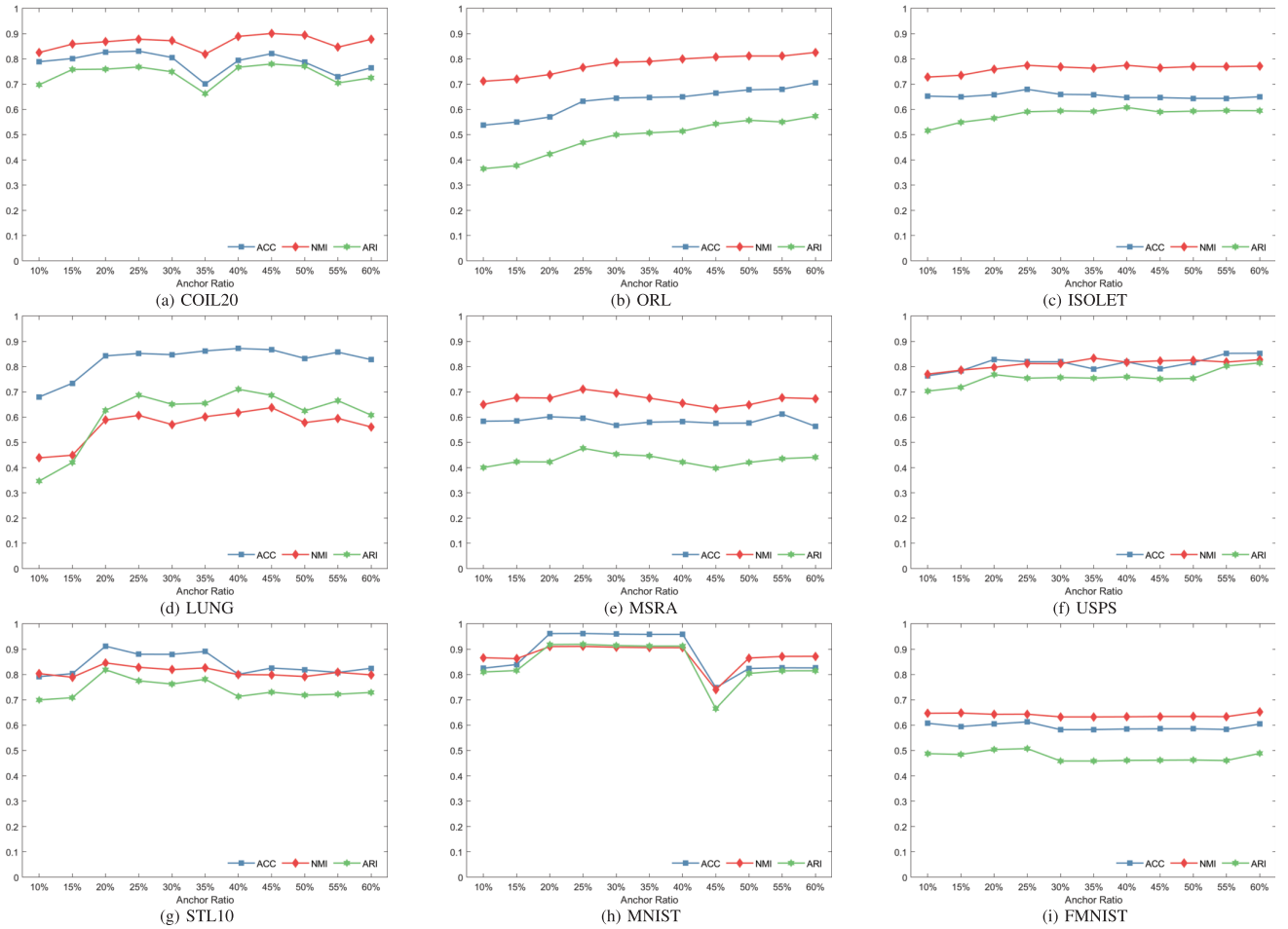


Fig. 5. Performance of 3SC under different numbers of anchors.

TABLE VI
ACCURACY OF 2SC AND 3SC UNDER DIFFERENT KERNELS

Method	Dataset	Gau.	Cau.	Lin.	Pol.	Sig.
2SC	COIL20	98.2	98.0	98.1	98.3	98.3
	ORL	71.8	71.3	72.3	70.0	70.0
	ISOLET	66.3	65.5	66.9	64.6	64.9
	LUNG	88.7	87.2	89.2	88.7	88.7
	MSRA	60.9	61.2	60.8	60.8	60.8
	USPS	85.0	83.8	90.5	85.0	83.0
3SC	COIL20	83.1	80.6	80.8	80.2	80.5
	ORL	63.3	62.5	61.5	60.8	59.3
	ISOLET	68.0	67.8	66.5	66.9	67.2
	LUNG	85.2	84.2	83.7	83.7	80.3
	MSRA	59.6	57.1	57.4	57.1	57.3
	USPS	78.5	77.6	85.7	77.8	81.4
	STL10	88.4	90.3	90.8	90.4	89.9
	MNIST	96.1	82.5	96.2	82.4	96.2
	FMNIST	61.3	59.6	60.2	58.8	58.1

performance among all compared algorithms. As shown in Table VIII, MV3SC ran successfully on three additional large datasets, achieving optimal performance in most cases, while LMVSC, TBGL, and OMSC encountered timeouts or memory shortages.

2) *Runtime*: Table VII shows that, MV2SC has a little time overhead, and MV3SC further reduces this time overhead. In terms of efficiency, both are not only great improvements to

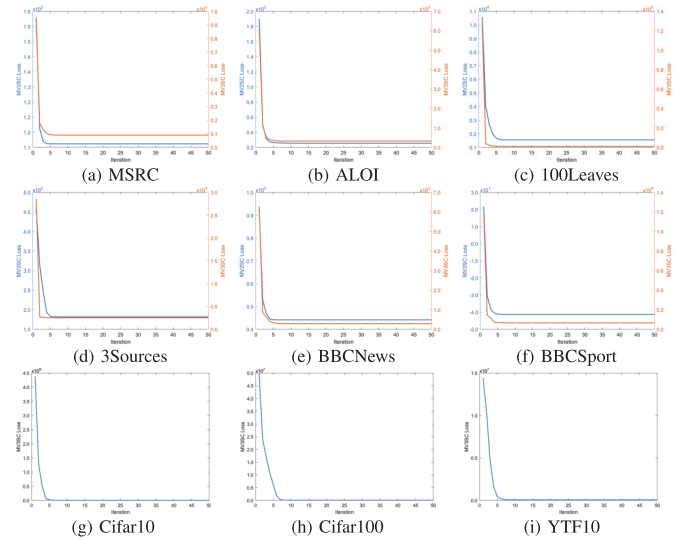


Fig. 6. Convergence curves of MV2SC and MV3SC during optimization.

MCLES, but also better than other algorithms. Furthermore, Table VIII shows that, MV3SC remains efficient when running on large-scale datasets.

3) *Convergence*: Figure 6 depicts the convergence curves. As the iterations increase, the losses of MV2SC and MV3SC

TABLE VII

ACCURACY (ACC% $\pm$ STD%), NORMALIZED MUTUAL INFORMATION (NMI% $\pm$ STD%), ADJUSTED RAND INDEX (ARI% $\pm$ STD%) AND RUNTIME (S) FOR MULTI-VIEW CLUSTERING ALGORITHM COMPARISON

Metric	ACC	NMI	ARI	Time	ACC	NMI	ARI	Time	ACC	NMI	ARI	Time
Dataset	MSRC				ALOI				100Leaves			
LMVSC	79.4 $\pm$ 1.0	65.9 $\pm$ 1.1	58.0 $\pm$ 1.6	1.3E+00	63.2 $\pm$ 4.1	64.7 $\pm$ 2.7	48.6 $\pm$ 6.1	1.9E+01	71.3 $\pm$ 0.7	86.4 $\pm$ 0.3	62.4 $\pm$ 1.0	5.9E+01
SFMC	60.0 $\pm$ 0.0	54.4 $\pm$ 0.0	39.2 $\pm$ 0.0	7.0E-01	77.0 $\pm$ 0.0	72.4 $\pm$ 0.0	52.6 $\pm$ 0.0	3.3E+00	57.6 $\pm$ 0.0	66.3 $\pm$ 0.0	09.1 $\pm$ 0.0	7.2E+00
TBGL	88.1$\pm$0.0	83.1$\pm$0.0	73.4 $\pm$ 0.0	1.2E+00	78.0 $\pm$ 0.0	71.3 $\pm$ 0.0	47.2 $\pm$ 0.0	5.2E+01	51.8 $\pm$ 0.0	52.5 $\pm$ 0.0	02.7 $\pm$ 0.0	1.1E+02
OMSC	59.5 $\pm$ 0.0	54.1 $\pm$ 0.0	44.9 $\pm$ 0.0	3.0E+00	66.4 $\pm$ 0.0	68.2 $\pm$ 0.0	57.5 $\pm$ 0.0	1.3E+01	37.7 $\pm$ 0.0	65.3 $\pm$ 0.0	19.3 $\pm$ 0.0	2.6E+01
CAMVC	69.5 $\pm$ 2.3	60.6 $\pm$ 1.6	50.1 $\pm$ 1.2	4.0E-02	56.0 $\pm$ 0.0	55.3 $\pm$ 0.1	42.9 $\pm$ 0.1	1.6E-01	69.7 $\pm$ 1.0	85.7 $\pm$ 0.2	60.7 $\pm$ 0.7	4.9E-01
SMVAGC	85.7 $\pm$ 0.0	75.8 $\pm$ 0.0	71.8 $\pm$ 0.0	1.1E+00	81.0 $\pm$ 0.0	81.9$\pm$0.0	73.1 $\pm$ 0.0	3.3E+00	59.0 $\pm$ 0.0	79.0 $\pm$ 0.0	47.3 $\pm$ 0.0	8.2E-01
MV3SC	88.1$\pm$0.0	77.7 $\pm$ 0.0	74.1$\pm$0.0	1.3E-01	88.2$\pm$0.0	80.7 $\pm$ 0.0	73.9$\pm$0.0	2.1E-01	87.5$\pm$1.0	93.5$\pm$0.2	81.6$\pm$0.6	8.2E-01
AMGL	77.1 $\pm$ 5.2	71.7 $\pm$ 2.9	62.9 $\pm$ 4.2	2.2E-01	62.7 $\pm$ 3.0	62.7 $\pm$ 0.6	40.5 $\pm$ 1.5	4.8E+00	82.3 $\pm$ 1.9	91.5 $\pm$ 1.9	57.8 $\pm$ 9.7	7.1E+00
SwMC	80.7 $\pm$ 5.7	76.3 $\pm$ 2.2	68.1 $\pm$ 3.8	3.8E-01	67.4 $\pm$ 1.2	65.8 $\pm$ 1.8	39.3 $\pm$ 2.9	1.2E+01	83.1 $\pm$ 6.2	90.4 $\pm$ 4.3	62.6 $\pm$ 9.6	7.7E+00
MLAN	85.7 $\pm$ 0.0	74.9 $\pm$ 0.0	70.4 $\pm$ 0.0	1.8E-01	76.6 $\pm$ 1.1	69.2 $\pm$ 1.1	49.5 $\pm$ 1.7	1.0E+00	88.2 $\pm$ 0.5	94.0 $\pm$ 0.1	80.7 $\pm$ 0.7	3.4E+00
GMC	81.9 $\pm$ 0.0	77.4 $\pm$ 0.0	68.2 $\pm$ 0.0	4.1E-01	66.6 $\pm$ 0.0	63.8 $\pm$ 0.0	36.4 $\pm$ 0.0	1.3E+00	88.4 $\pm$ 0.0	93.2 $\pm$ 0.0	61.6 $\pm$ 0.0	2.6E+00
MCGC	85.7 $\pm$ 0.0	76.2 $\pm$ 0.0	72.3 $\pm$ 0.0	1.7E-01	75.9 $\pm$ 0.0	74.8 $\pm$ 0.0	63.1 $\pm$ 0.0	5.6E+00	87.8 $\pm$ 0.0	93.7 $\pm$ 0.0	82.3 $\pm$ 0.0	1.7E+01
CDMGC	78.6 $\pm$ 0.8	75.3 $\pm$ 0.1	67.6 $\pm$ 0.0	2.9E-01	69.5 $\pm$ 0.1	68.7 $\pm$ 0.1	43.7 $\pm$ 0.2	7.1E+00	85.7 $\pm$ 0.7	93.3 $\pm$ 0.9	63.3 $\pm$ 7.4	1.4E+01
MCLES	83.1 $\pm$ 6.8	79.2 $\pm$ 2.6	71.6 $\pm$ 4.4	3.7E+00	48.6 $\pm$ 1.2	50.5 $\pm$ 1.1	34.4 $\pm$ 0.3	1.7E+03	81.0 $\pm$ 2.2	91.3 $\pm$ 1.3	68.0 $\pm$ 6.9	4.4E+04
MV2SC	93.8$\pm$0.0	87.8$\pm$0.0	85.9$\pm$0.0	1.4E-01	88.7$\pm$0.0	83.2$\pm$0.0	77.4$\pm$0.0	8.2E-01	92.4$\pm$0.9	96.5$\pm$0.3	89.0$\pm$0.8	4.4E+00
Dataset	3Sources				BBCNews				BBCSport			
LMVSC	58.8 $\pm$ 6.3	58.9 $\pm$ 4.1	44.5 $\pm$ 7.4	8.8E-01	63.3 $\pm$ 2.4	58.4 $\pm$ 3.5	46.1 $\pm$ 5.4	1.1E+01	66.4 $\pm$ 1.3	47.7 $\pm$ 1.7	32.5 $\pm$ 2.5	3.5E+00
SFMC	55.6 $\pm$ 0.0	34.9 $\pm$ 0.0	28.4 $\pm$ 0.0	4.9E-01	69.2 $\pm$ 0.0	47.1 $\pm$ 0.0	45.1 $\pm$ 0.0	2.1E+00	60.6 $\pm$ 0.0	39.8 $\pm$ 0.0	42.9 $\pm$ 0.0	1.1E+00
TBGL	43.2 $\pm$ 0.0	17.1 $\pm$ 0.0	01.0 $\pm$ 0.0	7.4E-01	56.2 $\pm$ 0.0	25.2 $\pm$ 0.0	14.9 $\pm$ 0.0	1.8E+01	82.6 $\pm$ 0.0	65.2 $\pm$ 0.0	68.3 $\pm$ 0.0	5.5E+00
OMSC	42.8 $\pm$ 0.0	17.5 $\pm$ 0.0	12.9 $\pm$ 0.0	4.0E+00	31.4 $\pm$ 0.0	14.5 $\pm$ 0.0	05.9 $\pm$ 0.0	6.6E+01	36.5 $\pm$ 0.0	08.5 $\pm$ 0.0	09.2 $\pm$ 0.0	1.3E+01
CAMVC	54.9 $\pm$ 3.1	53.0 $\pm$ 1.5	34.9 $\pm$ 3.1	2.3E-01	60.4 $\pm$ 2.4	55.0 $\pm$ 3.5	43.7 $\pm$ 5.4	1.8E+00	69.6 $\pm$ 6.4	56.5 $\pm$ 5.3	50.5 $\pm$ 6.7	5.2E-01
SMVAGC	55.0 $\pm$ 0.0	54.9 $\pm$ 0.0	39.9 $\pm$ 0.0	2.2E+00	54.2 $\pm$ 0.0	31.8 $\pm$ 0.0	16.1 $\pm$ 0.0	2.2E+01	57.9 $\pm$ 0.0	37.3 $\pm$ 0.0	25.5 $\pm$ 0.0	6.5E+00
MV3SC	80.9$\pm$0.7	65.0$\pm$0.6	64.0$\pm$1.1	1.2E-01	90.7$\pm$0.0	74.1$\pm$0.0	77.9$\pm$0.0	7.0E-02	96.3$\pm$0.0	88.0$\pm$0.0	91.3$\pm$0.0	1.2E-01
AMGL	68.9 $\pm$ 3.0	52.9 $\pm$ 4.2	42.9 $\pm$ 4.7	1.1E-01	80.6 $\pm$ 7.6	61.4 $\pm$ 9.2	62.6 $\pm$ 9.9	1.4E+00	85.5 $\pm$ 7.4	80.2 $\pm$ 8.0	77.5 $\pm$ 9.3	6.9E-01
SwMC	70.9 $\pm$ 4.2	56.1 $\pm$ 4.1	46.2 $\pm$ 6.0	4.4E-01	69.5 $\pm$ 0.7	48.1 $\pm$ 0.9	46.5 $\pm$ 1.4	4.3E+00	90.4 $\pm$ 7.1	83.6 $\pm$ 8.1	82.1 $\pm$ 9.5	4.1E+00
MLAN	76.2 $\pm$ 0.3	67.0 $\pm$ 0.0	57.9 $\pm$ 0.3	1.4E-01	83.7 $\pm$ 0.0	64.9 $\pm$ 0.0	68.0 $\pm$ 0.0	1.2E+00	97.1 $\pm$ 0.0	90.0 $\pm$ 0.0	92.1 $\pm$ 0.0	6.6E-01
GMC	69.2 $\pm$ 0.0	54.8 $\pm$ 0.0	44.3 $\pm$ 0.0	3.4E-01	69.1 $\pm$ 0.0	48.1 $\pm$ 0.0	47.6 $\pm$ 0.0	2.0E+00	97.8 $\pm$ 0.0	92.6 $\pm$ 0.0	93.4 $\pm$ 0.0	7.7E-01
MCGC	66.9 $\pm$ 0.0	63.9 $\pm$ 0.0	55.4 $\pm$ 0.0	1.2E-01	76.9 $\pm$ 0.0	61.2 $\pm$ 0.0	52.9 $\pm$ 0.0	1.8E+00	88.8 $\pm$ 0.0	81.0 $\pm$ 0.0	82.4 $\pm$ 0.0	8.2E-01
CDMGC	64.7 $\pm$ 1.9	47.8 $\pm$ 1.5	34.5 $\pm$ 1.8	2.6E-01	72.7 $\pm$ 0.1	55.0 $\pm$ 0.2	53.4 $\pm$ 0.1	2.9E+00	73.6 $\pm$ 0.1	69.5 $\pm$ 0.4	59.3 $\pm$ 0.3	1.1E+00
MCLES	72.9 $\pm$ 2.3	57.8 $\pm$ 3.7	54.7 $\pm$ 4.1	1.8E+00	71.0 $\pm$ 1.2	48.7 $\pm$ 1.6	48.4 $\pm$ 2.5	2.5E+02	82.6 $\pm$ 7.0	74.6 $\pm$ 8.2	75.2 $\pm$ 9.3	9.2E+01
MV2SC	76.3$\pm$0.0	70.4$\pm$0.0	66.1$\pm$0.0	1.3E-01	91.2$\pm$0.0	76.6$\pm$0.2	80.6$\pm$0.1	2.6E-01	98.2$\pm$0.0	93.3$\pm$0.0	95.2$\pm$0.0	1.7E-01

TABLE VIII

ACCURACY (ACC% $\pm$ STD%), NORMALIZED MUTUAL INFORMATION (NMI% $\pm$ STD%), ADJUSTED RAND INDEX (ARI% $\pm$ STD%) AND RUNTIME (S) FOR EXTRA NEAR-LINEAR MULTI-VIEW CLUSTERING ALGORITHM COMPARISON

Metric	ACC	NMI	ARI	Time	ACC	NMI	ARI	Time	ACC	NMI	ARI	Time
Dataset	Cifar10				Cifar100				YTF10			
LMVSC	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO	TO
SFMC	89.5 $\pm$ 0.0	92.0 $\pm$ 0.0	88.5 $\pm$ 0.0	1.1E+04	99.9$\pm$0.0	99.9$\pm$0.0	99.9$\pm$0.0	9.0E+03	62.3 $\pm$ 0.0	62.4 $\pm$ 0.0	43.1 $\pm$ 0.0	4.1E+03
TBGL	OM	OM	OM	OM	OM	OM	OM	OM	OM	OM	OM	OM
OMSC	OM	OM	OM	OM	OM	OM	OM	OM	TO	TO	TO	TO
CAMVC	99.4$\pm$0.0	98.3$\pm$0.0	98.7$\pm$0.0	2.5E+03	89.7 $\pm$ 0.0	97.9 $\pm$ 0.0	91.6 $\pm$ 0.0	3.3E+03	56.1 $\pm$ 0.0	60.3 $\pm$ 0.0	38.3 $\pm$ 0.0	1.4E+03
SMVAGC	95.5 $\pm$ 0.0	89.4 $\pm$ 0.0	90.3 $\pm$ 0.0	2.9E+04	94.6 $\pm$ 0.0	98.8 $\pm$ 0.0	95.2 $\pm$ 0.0	4.2E+04	77.6 $\pm$ 0.0	79.4 $\pm$ 0.0	69.2 $\pm$ 0.0	2.1E+04
MV3SC	99.1 $\pm$ 0.0	97.3 $\pm$ 0.0	97.9 $\pm$ 0.0	4.5E+01	99.9$\pm$0.0	99.9$\pm$0.0	99.9$\pm$0.0	2.3E+02	78.3$\pm$0.0	82.9$\pm$0.0	74.2$\pm$0.0	1.3E+02

TO: Time Out. OM: Out of Memory.

first drop sharply and then level off. This implies that our models are able to converge rapidly and stably.

4) *Neighbor Number*: Take 1, 2, $\dots$, 16 neighbor numbers in turn to observe the accuracy change trends of MV2SC and MV3SC with Gaussian/linear kernel (MV2SC_G/L and MV3SC_G/L). As captured in Figure 7, the increase in neighbor numbers leads to an initial increase in accuracy and then a gradual flattening in most cases. Slightly different from the single-view cases, flattening comes later, but still only a small number of neighbors is required. Although the accuracy of MV3SC fluctuates up and down with the increase

in neighbor numbers when running on YTF10, fortunately, it achieves the best performance with just six neighbors. Therefore, our model parameters remain easy to set.

5) *Anchor Ratio*: Take 10%, 15%, $\dots$, 60% samples as anchors in turn to observe the performance change trend of MV3SC. As captured in Figure 8, similar to the single-view cases, an excessive number of anchors fails to bring significant performance improvement. This implies that our multi-view model is lightweight and requires only a small number of anchors. Of course, larger datasets typically require a greater number of anchors.

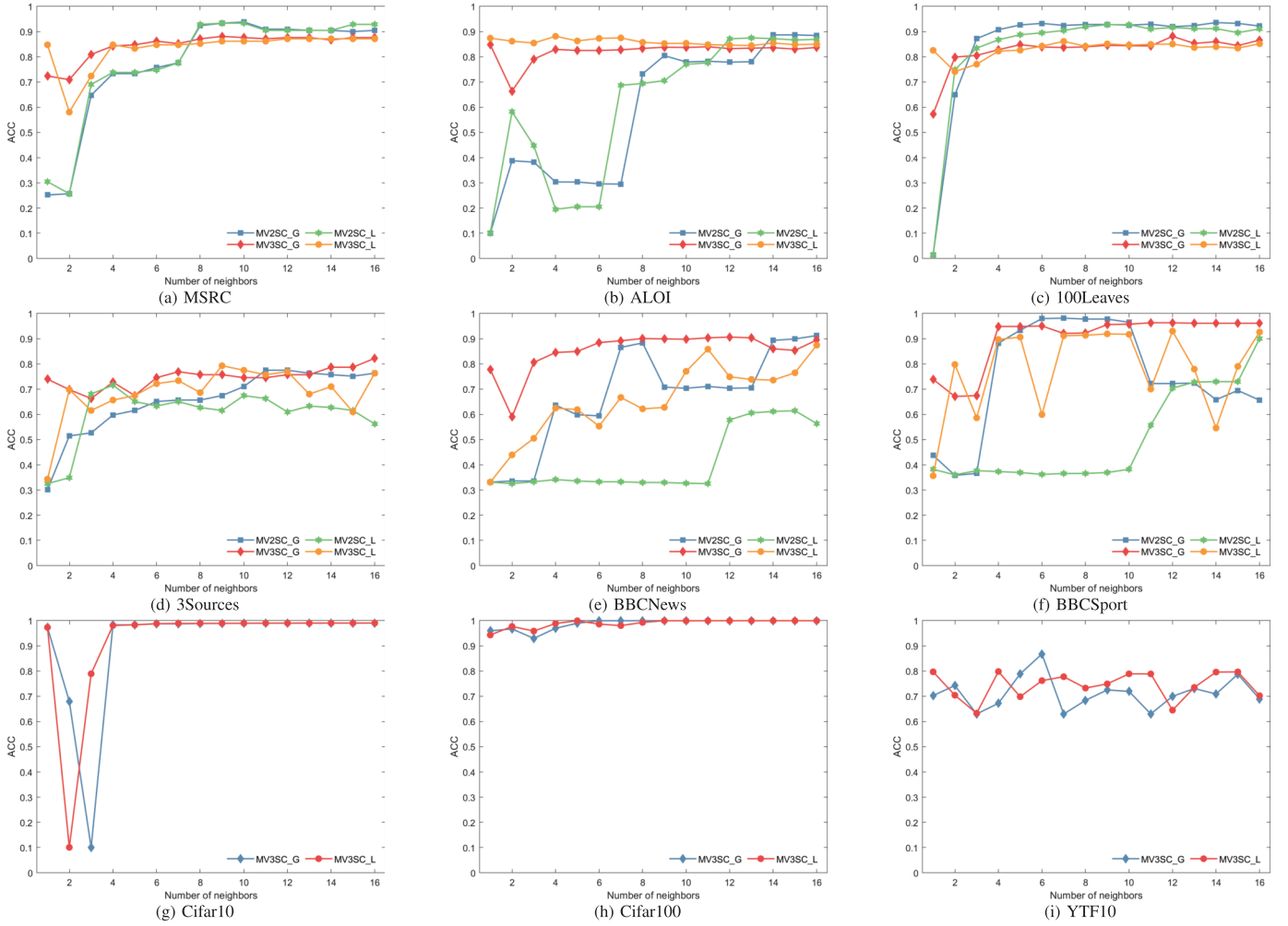


Fig. 7. Accuracy of MV2SC and MV3SC under different numbers of neighbors.

TABLE IX

ACCURACY OF MV3SC UNDER DIFFERENT ANCHOR STRATEGIES

Dataset	Random	K-means	VDA
MSRC	79.5	86.7	89.1
ALOI	83.9	89.2	89.6
100Leaves	81.1	90.8	87.9
3Sources	65.7	77.5	80.5
BBCNews	81.8	87.6	90.1
BBCSport	90.3	95.6	96.3
Cifar10	97.3	99.0	99.0
Cifar100	99.9	99.9	99.9
YTF10	69.8	80.0	78.6

6) *Anchor Selection*: Several anchor selection strategies are tested for MV3SC, including random sampling, k-means sampling, and VDA sampling. As described in Table IX, the situation is similar for MV3SC and 3SC: random sampling is not recommended, while VDA sampling and k-means sampling are more suitable for cases with a large and small number of anchors, respectively.

7) *Kernel Type*: More types of kernels are tested, including Gaussian, Cauchy, linear, polynomial and sigmoid kernels. As reported in Table X, unlike the single-view cases, both MV2SC

TABLE X

ACCURACY OF MV2SC AND MV3SC UNDER DIFFERENT KERNELS

Method	Dataset	Gau.	Cau.	Lin.	Pol.	Sig.
MV2SC	MSRC	93.8	93.3	93.3	93.3	93.3
	ALOI	88.7	91.4	87.5	88.2	86.6
	100Leaves	93.6	94.5	92.8	92.4	85.9
	3Sources	77.5	78.7	71.6	62.1	71.0
	BBCNews	91.2	89.1	61.5	39.3	65.4
	BBCSport	98.2	97.4	90.1	48.0	87.5
MV3SC	MSRC	88.1	87.6	87.1	87.1	87.6
	ALOI	84.8	85.7	88.2	87.6	87.8
	100Leaves	88.1	87.9	86.1	88.1	86.6
	3Sources	82.3	81.1	79.3	70.4	75.2
	BBCNews	90.7	90.2	87.5	87.5	88.3
	BBCSport	96.3	96.3	93.0	80.2	91.4
	Cifar10	99.0	99.1	99.1	99.0	99.0
	Cifar100	99.9	99.9	99.9	99.9	99.9
	YTF10	77.3	77.4	78.7	79.0	77.4

and MV3SC almost always achieve the best performance when equipped with Gaussian kernel, and occasionally when equipped with Cauchy kernel or linear kernel. Therefore, Gaussian kernel may be a suitable recommendation. But there is still no need to spend too much effort on kernel selection.

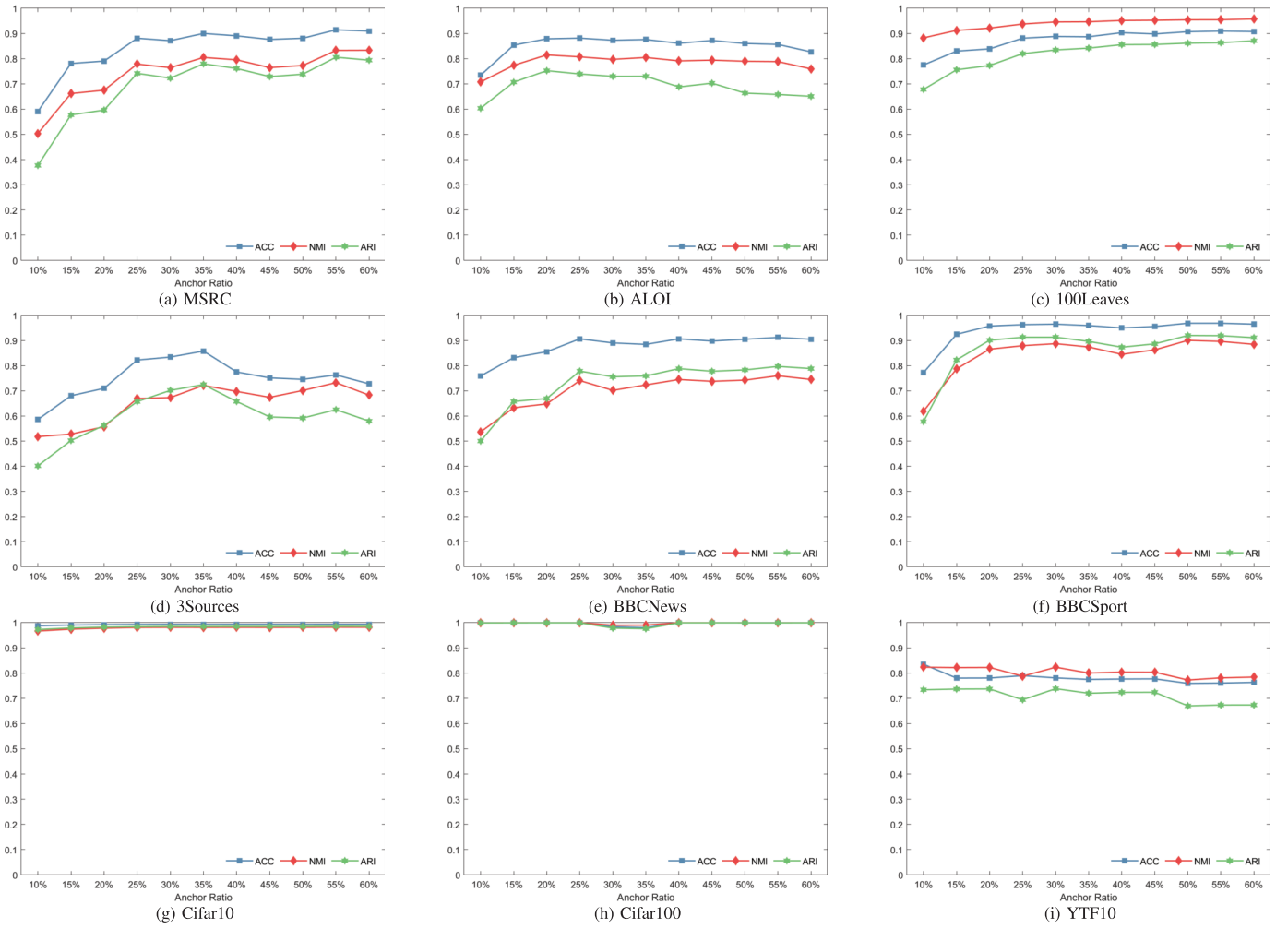


Fig. 8. Performance of MV3SC under different numbers of anchors.

TABLE XI

ACCURACY (ACC%±STD%), NORMALIZED MUTUAL INFORMATION (NMI%±STD%), ADJUSTED RAND INDEX (ARI%±STD%) AND RUNTIME (S) FOR MULTI-KERNEL CLUSTERING ALGORITHM COMPARISON

Metric	ACC	NMI	ARI	Time	ACC	NMI	ARI	Time	ACC	NMI	ARI	Time
Dataset	MSRC				ALOI				100Leaves			
MKKM	70.1±0.9	56.1±1.5	48.1±1.7	2.2E-02	61.9±1.2	64.3±0.5	52.0±0.8	6.4E-01	49.4±0.6	74.6±0.3	38.6±0.9	7.8E+00
LMKKM	65.2±0.0	53.0±0.0	41.7±0.0	2.5E+00	52.9±0.0	51.2±0.0	35.2±0.0	8.9E+01	48.6±0.0	73.4±0.0	36.3±0.0	1.6E+02
ONKC	80.5±1.4	68.4±1.4	62.2±2.0	4.2E-02	74.9±0.2	71.4±0.6	61.8±0.2	1.1E+00	77.3±2.3	89.9±0.8	70.6±2.3	3.0E+00
HMKC	85.7±0.0	74.4±0.0	70.5±0.0	2.4E-01	55.0±0.0	59.2±0.0	41.0±0.0	7.5E+00	85.3±0.0	94.3±0.0	81.9±0.0	2.0E+01
DPMKKM	88.6±0.0	80.6±0.0	76.2±0.0	2.7E-02	63.9±0.0	60.2±0.0	46.7±0.0	1.5E-01	76.9±0.0	88.9±0.0	68.1±0.0	4.4E-01
SMKKM	81.1±3.4	68.8±3.5	63.1±4.9	2.9E-01	70.1±5.7	70.7±1.5	60.3±3.8	5.1E+00	76.9±1.7	89.9±0.5	71.0±1.3	9.6E+00
LSMKKM	74.3±0.9	65.7±1.3	58.3±1.1	2.1E-01	66.6±0.2	65.4±0.3	52.9±0.2	5.0E+00	83.7±1.2	94.0±0.3	80.6±1.0	1.5E+01
MV3SC	93.8±0.0	87.8±0.0	85.9±0.0	1.3E-01	88.7±0.0	83.2±0.0	77.4±0.0	2.1E-01	92.4±0.9	96.5±0.3	89.0±0.8	8.2E-01
Dataset	3Sources				BBCNews				BBCSport			
MKKM	57.5±1.8	60.3±0.9	45.2±1.9	1.8E-02	76.9±0.1	59.1±0.2	53.0±0.2	1.8E-01	89.9±0.0	80.8±0.0	84.1±0.0	5.7E-02
LMKKM	60.4±0.0	61.4±0.0	49.3±0.0	3.8E-01	76.6±0.0	58.9±0.0	52.6±0.0	2.4E+01	89.9±0.0	80.8±0.0	84.1±0.0	3.3E+00
ONKC	54.4±1.1	59.6±2.1	43.0±2.7	2.8E-02	63.0±0.1	61.3±0.0	51.9±0.0	5.0E-01	78.5±1.6	78.0±1.9	76.7±0.6	4.4E-01
HMKC	61.5±0.0	54.0±0.0	41.9±0.0	1.2E-01	75.5±0.0	59.2±0.0	49.4±0.0	5.6E-01	87.3±0.0	76.9±0.0	80.5±0.0	3.4E-01
DPMKKM	62.7±0.0	61.0±0.0	49.9±0.0	1.9E-02	72.7±0.0	55.1±0.0	44.3±0.0	1.1E-01	94.3±0.0	84.2±0.0	86.4±0.0	1.5E-01
SMKKM	57.2±1.8	58.5±3.6	43.6±3.7	1.4E-01	76.9±0.1	59.0±0.2	53.0±0.1	9.8E-01	89.9±0.1	80.8±0.1	84.1±0.0	4.6E-01
LSMKKM	71.4±2.0	59.1±1.2	53.7±2.2	1.2E-01	64.5±0.1	54.1±0.1	49.1±0.2	1.1E+00	73.3±0.1	71.2±0.3	64.6±0.2	4.3E-01
MV3SC	76.3±0.0	70.4±0.0	66.1±0.0	1.2E-01	91.2±0.0	76.6±0.2	80.6±0.1	7.2E-02	98.2±0.0	93.3±0.0	95.2±0.0	1.2E-01

D. Multi-Kernel Clustering

Considering the connection between our models and kernel clustering, we further compare MV2SC with various

state-of-the-art multi-kernel k-means algorithms. Using the features of each view in the multi-view datasets, multiple Gaussian kernels are constructed as inputs to multi-kernel

clustering. As shown in Table XI, our model has indelible advantages in ACC, NMI, and ARI. It is a pity that our operating efficiency will be relatively low. Fortunately, the efficiency gap is not significant, but the effectiveness is very significant.

VI. CONCLUSION

In this paper, we propose 3SC for single-view scalable subspace clustering. Under constraints and algebraic operations, 3SC becomes a simple approximation of scalable subspace clustering. It allows for a more user-friendly optimization technique, with linear complexity regarding both the number of samples and anchors, further enhancing scalability and providing efficient operations. Furthermore, our separate fusion strategy extends 3SC into MV3SC for multi-view scalable subspace clustering. This strategy divides the task into two convex sub-problems, thereby escaping the cumbersomeness and non-convexity caused by multi-variable alternating iterations. Surprisingly, this strategy is more interpretable than the conventional self-weighting methods, but does not significantly increase the burden of storage and calculation. Finally, comprehensive experiments show that our models are still very competitive in terms of performance while ensuring efficiency.

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Zhounin Lu received the M.S. degree in computer technology from Fuzhou University, China, in 2021. He is currently pursuing the Ph.D. degree with the School of Computer Science, Northwestern Polytechnical University. His research interests include representation learning, multi-view learning, graph learning, federated learning, and their applications, such as clustering and anomaly detection.



Feiping Nie (Senior Member, IEEE) received the Ph.D. degree in computer science from Tsinghua University, China, in 2009. He is currently a Full Professor at Northwestern Polytechnical University, China. His research interests include machine learning and its applications, such as pattern recognition, data mining, computer vision, image processing, and information retrieval. He has published more than 100 articles in the following journals and conferences, such as IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, IJCV, IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON KNOWLEDGE AND DATA ENGINEERING, ICML, NIPS, KDD, IJCAI, AAAI, ICCV, CVPR, and ACM MM. His articles have been cited more than 20 000 times and the H-index is 84. He is serving as an associate editor or a PC member for several prestigious journals and conferences in the related fields.



Linru Ma received the B.E. and Ph.D. degrees in communication and information systems from the National University of Defense Technology, Changsha, China, in 2000 and 2007, respectively. She is currently a Researcher at the Institute of Data and Intelligence and a Professor at the School of Systems Science and Engineering, Sun Yat-sen University. Her research mainly focuses on pattern recognition, machine learning, and data mining on anomaly detection. Her current research interests include resilience intelligent uncrewed systems.



Rong Wang received the B.S. degree in information engineering, the M.S. degree in signal and information processing, and the Ph.D. degree in computer science from Xi'an Research Institute of Hi-Tech, Xi'an, China, in 2004, 2007, and 2013, respectively. From 2007 to 2013, he also studied at the Department of Automation, Tsinghua University, Beijing, China, for his Ph.D. degree. He is currently an Associate Professor at the School of Artificial Intelligence, Optics and ElectroNics (iOPEN), Northwestern Polytechnical University, Xi'an. His research interests focus on machine learning and its applications.

Xuelong Li (Fellow, IEEE) is the CTO and Chief Scientist of China Telecom, where he founded the Institute of Artificial Intelligence (TeleAI) of China Telecom.